

CAEN RFID Proton
(R4320P)

The Proton (R4320P) is a rugged long-range RAIN RFID reader from the easy2read product line, designed for industrial environment installations. With 4 antenna ports, IP65 protection, PoE, and M12 industrial connectors, it is an ideal choice for industrial automation, Industry 4.0 solutions, and outdoor RAIN RFID portals.



RFID

Standard Compliance	EPC Class 1 Gen 2 - ISO 18000-63
Frequency Range	865.600÷867.600 MHz (ETSI EN 302 208 v. 3.1.1); 902÷928 MHz (FCC part 15.247)
RF Power	Up to 31.5 dBm (1.4 W) conducted (ETSI); Up to 30 dBm (1 W) conducted (FCC)
RX Sensitivity	-84 dBm – 10% PER, assuming 20 dB antenna RL @ 31.5 dBm output
Number of Channels	4 channels (ETSI); 50 hopping channels (FCC)
Antenna Connectors	4 TNC Reverse Polarity
Receiving Capability	Gen 2 Dense Reader Mode Management; Data rate up to 400 kbit/s

PLATFORM

CPU	ARM9 @ 400 MHz on Atmel AT91SAM9G25
Operating System	Linux

CONNECTIVITY

Serial	RS232 on M12, up to 115.2 kbit/s
Ethernet	10/100/1000Base-T on M12
PoE	IEEE 802.3af
I/O Interface	M12; 2 optically isolated inputs; 2 photorelay outputs (500 mA max)

MECHANICAL & POWER

Dimensions	(W) 131 x (L) 106 x (H) 50 mm
Weight	530 g
Power Supply	9÷36 V DC power supply (12 W); PoE IEEE 802.3af (12.95 W)
Status Indicators	Multicolour LEDs: Power, Activity, Status and Applications

ENVIRONMENT

IP Rating	IP65
Operating Temperature	-10 °C to +55 °C

FEATURE HIGHLIGHTS

- RAIN RFID (UHF EPC Class1 Gen2, ISO 18000-63) compliant
- Multiregional support
- Four 50 Ohm TNC-RP antenna connectors
- Power over Ethernet interface
- Up to 31.5 dBm (1.4 W) output power
- Internal scripting engine
- IP65 in compact form factor
- M12 industrial connectors

PRODUCT GROUP

Category	RFID Reader
Product Group	Industrial Long Range RAIN RFID Reader